Year 12 Maths - Exponential models

Similar questions - Exponential growth and decay

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A radioactive sample has mass m grams at time t minutes, modelled by $m = 200\mathrm{e}^{-kt}$, where k is a positive constant. The mass halves every 20 minutes.

(a) Show that $k = \dfrac{\ln 2}{20}$. **[3]**

(b) Find the mass of the sample after 50 minutes. **[2]**

(c) Find the rate at which the mass is decreasing when $t = 20$. **[3]**

(d) Find the time taken for the mass to fall below 1 g. **[3]**

2 The populations of two species of fish in a lake are modelled by $A = 500 \times 1.04^t$ and $B = 900 \times 0.97^t$, where t is the number of years from the start of the study.

(a) Find the population of each species when $t = 10$. **[2]**

(b) Find the number of years it takes for species A to first exceed twice its initial population. **[3]**

(c) Show that the two populations are equal when $t = \dfrac{\ln 1.8}{\ln 1.04 - \ln 0.97}$ and evaluate this to 3 significant figures. **[4]**

(d) Describe what happens to each population in the long term, and comment on whether the models are realistic. **[2]**

Question 1

(a) $100 = 200e^{-20k}$, so $e^{-20k} = \frac{1}{2}$, $-20k = -\ln 2$ and $k = \dfrac{\ln 2}{20}$

(b) $m = 200e^{-2.5\ln 2} = 200 \times 2^{-2.5} = 35.4$ g

(c) $\dfrac{dm}{dt} = -km$. At $t = 20$, $m = 100$, so the rate is $-100k = -3.47$ g per minute, i.e. decreasing at 3.47 g/min

(d) $200e^{-kt} < 1$, so $e^{-kt} < \frac{1}{200}$ and $t > \dfrac{\ln 200}{k} = 153$ minutes

Question 2

(a) $A = 500 \times 1.04^{10} = 740$, $B = 900 \times 0.97^{10} = 664$ (to 3 s.f.)

(b) $500 \times 1.04^t > 1000$, so $1.04^t > 2$ and $t > \dfrac{\ln 2}{\ln 1.04} = 17.67$. It takes 18 years

(c) $500 \times 1.04^t = 900 \times 0.97^t$, so $\left(\dfrac{1.04}{0.97}\right)^t = 1.8$ and $t = \dfrac{\ln 1.8}{\ln 1.04 - \ln 0.97} = 8.43$ years

(d) A grows without limit and B tends to 0. A real lake can only support a limited number of fish, so A cannot keep growing; B would not decrease smoothly to a fraction of a fish